RELATIVE FINITE PRESENTATION SATISFIES THE TWO-OUT-OF-THREE RULE ON SHORT EXACT SEQUENCES IN THE OUTER TERMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16536-16548 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 16550-16555, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Translation-protocol template (same family as collected L16133); sequenced after its dictionary card L16180 in this same wave.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma ses-relatively-finite-presentation; the relative-finite-presentation dictionary (sibling card L16180).

Source reference. Original lemma 0671: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16555–16555. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be a short exact sequence of quasi-coherent $\mathcal{O}_X$-modules.*

(1) *If $\mathcal{F}', \mathcal{F}''$ are finitely presented relative to S, then so is $\mathcal{F}$.*
(2) *If $\mathcal{F}'$ is a finite type $\mathcal{O}_X$-module and $\mathcal{F}$ is finitely presented relative to S, then $\mathcal{F}''$ is finitely presented relative to S.*

Proof. Translation of the result of More on Algebra, Lemma 0671 into the language of schemes. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.